

Algorithms That Choose Us

Participant Material · [Anonymized Institution]

Mariana's Internship

Recommendation Systems, Algorithmic Biases, and the Construction of Worldview

Who is Mariana?

Mariana Santos, 20 years old — A third-year student in the Systems Development Technical Course at a [Anonymized Federal Institute]. A good student. Curious. Wants to work in the technology field.

Over the last two years, Mariana watched hundreds of hours of YouTube videos about programming, startups, and the tech job market. The YouTube algorithm acted as a sort of career advisor for her — always suggesting the next video, the next channel, the next idea.

The Problem Begins Here

Mariana secured an internship at a recruitment startup that uses artificial intelligence to screen job candidates. On her first day, the technical team presented the company's job recommendation system and asked each intern to critically evaluate the tool.

Mariana's analysis surprised her supervisor—negatively. She submitted a report stating that the system was working correctly and that the recommendation results were fair and accurate.

When questioned about why she believed this, she replied:

"Algorithms are pure mathematics. They have no bias — they process data and deliver the best result. If the system recommends more men for leadership positions, it must be because they have more relevant profiles in the data."

She had no baseline or studies on bias in recommendation systems.

YouTube did not censor Mariana. It did not forbid her from searching for other content. But the recommendation algorithm did something much more subtle: over hundreds of interactions, it learned what kept her engaged and progressively narrowed the universe of voices, perspectives, and ideas that reached her — all while Mariana believed she was becoming increasingly well-informed.

This is a **filter bubble** — and it is not visible from the inside. Mariana did not know what she did not know. And it was precisely this lack of awareness that led her to analyze a discriminatory system as if it were fair.

Now It's Your Turn

1. How did the YouTube algorithm construct Mariana's worldview without her realizing it?
2. Does the startup's job recommendation system have a technical problem, an ethical problem, or both? What is the difference?
3. If Mariana's feed had been different — with more voices, perspectives, and debates —

would she have reached the same conclusion in her report?

Do not answer just yet. First, listen to what your peers think. Note down the reactions that emerge on the board — they will be the starting point for everything that follows.